

DECODED EXPLAINED

20th July, 2026

Finning: Cruel, Wasteful Practice

Finning: cruel, wasteful practice

Agence France Presse

Finning, the practice of removing a shark's fins and casting the rest of the shark overboard, often alive. The animal will subsequently drown. Finning is largely prohibited across a variety of regulatory zones, countries and international conventions, even if fishing certain shark species is generally allowed.

Yet in recent years, illegal shark fins have been repeatedly landed in Senegal's capital Dakar, one of Africa's busiest fishing ports, by dozens of Chinese and Taiwanese longline tuna vessels, according to a report released last week by the London-based Environmental Justice Foundation NGO.

The fins are then shipped to Asia for use in shark fin soup, a delicacy in certain countries, or in traditional medicine.

The practice is inherently cruel and wasteful, conservationists say, likening it to killing a rhinoceros for its horn. But brutality is not the only problem.

Global shark populations have been in rapid decline due to industrial fishing, with a study published in the journal Nature showing their population has plummeted, alongside that of their close relatives rays, 71% since 1970.

Finning also often occurs in tandem with other types of illegal, unreported, and unregulated fishing and human



A picture taken on November 17, 2022, shows shark fins laid out on a tarp outside a shop in Hong Kong. AFP

rights abuses.

Senegal is a party to the International Commission for the Conservation of Atlantic Tunas, which regulates fishing operations related to tuna and tuna-like species in the Atlantic Ocean. It also bans the retention of certain species of sharks and specifies that fins cannot total more than 5% of the overall weight of sharks onboard. The rule is meant to prevent finning, but makes things difficult to quantify.



For feedback and suggestions

for 'Science', please write to science@thehindu.co.in with the subject 'Daily page'

Finning: Cruel, Wasteful Practice

Concept

- The Practice

- Finning involves removing a shark's fins and casting the rest of the shark overboard, often alive, causing it to subsequently drown

- Recent Case

- Illegal shark fins have been repeatedly landed in Senegal's capital Dakar, one of Africa's busiest fishing ports, by dozens of Chinese and Taiwanese longline tuna vessels, per a report by the Environmental Justice Foundation

- Regulatory Framework

- Senegal is party to the International Commission for the Conservation of Atlantic Tunas (ICCAT), which bans retention of certain shark species and caps fins at 5% of overall shark weight onboard hard to enforce in practice

Finning: Cruel, Wasteful Practice

- **Why Sharks Matter Ecologically**

- As apex predators, sharks regulate marine food webs and maintain ocean ecosystem balance; their slow reproduction rates make populations highly vulnerable to overfishing

- **What Is IUU Fishing?**

- Illegal, Unreported and Unregulated fishing activity that violates laws, goes unreported to authorities, or occurs outside regulatory frameworks, a major driver of global fisheries depletion

Shah Unveils India's First Immersive Language Museum

Shah unveils India's first immersive language museum

The Hindu Bureau
KOLKATA

Highlighting that a country's language carries its heritage and offers a window into its culture, Union Home Minister Amit Shah on Sunday inaugurated the Museum of Word at the National Library campus here and called it a step to preserve the *virasat* of India.

Developed by the National Library under the aegis of the Union Culture Ministry, the first immersive language museum in the country, also known as *Shabdlok*, celebrates India's rich linguistic heritage and showcases the evolution of its oral and written traditions through interactive and technology-driven exhibits.

"Our Prime Minister [Narendra Modi] has prioritised both *Vikas* (development) and *virasat* (heritage). This museum is a step to preserve India's rich linguistic and cultural heritage. Our language carries our heritage, and through languages you can understand a country's culture," Mr. Shah said addressing the inauguration event.

The museum features nine galleries exploring different facets of the history of languages and their influence on India's civilisa-



Home Minister Amit Shah at an event in Kolkata on Sunday. PTI

tion and culture. It seeks to present language as a living entity that has continuously shaped society while also being shaped by it.

Linguistic diversity

The first phase of the museum chronicles the diversity of Indian languages, scripts, and literature.

Rather than a conventional repository, it has been designed as an immersive cultural experience using digital displays, holograms, and motion-sensor technology to engage visitors.

Shabdlok celebrates India's 22 official languages and traces their journey from oral traditions and ancient manuscripts to printed works and modern electronic texts, highlighting the country's enduring linguistic diversity.

Shah Unveils India's First Immersive Language Museum

Concept

- **The Inauguration**

- Union Home Minister Amit Shah inaugurated the **Museum of Word (ShabdaloK)** at the National Library campus in Kolkata, calling it a step to preserve India's virasat (heritage)

- **Development and Purpose**

- Developed by the **National Library under the Union Culture Ministry, it is India's first immersive language museum,** celebrating linguistic heritage through interactive, technology-driven exhibits
- The first phase chronicles the diversity of Indian languages, scripts and literature; ShabdaloK celebrates India's 22 official languages, tracing their journey from oral traditions to modern electronic texts

- **India's 22 Scheduled Languages**

- Listed in the Eighth Schedule of the Constitution, these are languages officially recognised for purposes including translation of official documents and competitive exams

India's F&O Boom Needs Adequate Protections

India's F&O boom needs adequate protections

Headlines continue to highlight the devastating financial ruin of investors who trade in the **Futures and Options (F&O)** segment, as many of them get caught in a spiral of debt. Mounting financial pressure from these trading losses has recently led to multiple cases of suicide.

At their core, F&O contracts are derivatives. They allow investors to buy, sell, or lock in the price of an underlying asset such as a stock, index, or commodity at a future date without actually owning it. While institutions use these instruments to hedge existing portfolios against market downturns, many retail participants are drawn to them solely to speculate, lured by the promise of quick money.

According to SEBI reports, the average investor driving this frenzy is overwhelmingly male (86.3%), hails from small tier-2 and tier-3 cities (over 72%) and earns less than ₹5 lakh per annum (75%). A July 2025 SEBI study revealed that over 90% of individual traders incurred heavy losses over the four-year period from FY22; aggregate retail losses surpassed ₹2.8 lakh crore, with the average net loss per participant pegged above ₹4 lakh. Only 1% of retail traders managed a profit exceeding ₹1 lakh.

Rigged playing field

Over the last eight years, India has evolved from a marginal player into one of the world's largest F&O markets. This hyper-growth began during the FY16 to FY19 period, when exchanges introduced weekly-expiring contracts in addition to the traditional month-end expiries. Fuelled by a booming bull market, smooth mobile trading apps, instant UPI fund transfers, aggressive 'how-to-trade' content via social media, and cheap discount brokerages, entering the market became effortless. However, the psychological thrill of fast cash has blinded users to reality: over 75% of loss-making retail traders



Swetha Sekar

VP-Equity and Credit Research, Aionion Investment Managers LLP Chennai



Anand Srinivasan

Deep value investor in stocks and commodities

Mature derivatives markets do not rely on warnings; they enforce strict entry barriers and suitability systems to limit speculative risks

continue to trade year after year, even after sustaining heavy losses.

Retail traders lose because they are bringing knives to a gunfight. SEBI's research reveals that 97% of institutional profits and 96% of proprietary trading profits in the Indian F&O market are generated by algorithmic trading. These institutional desks utilise lightning-fast code co-located directly inside exchange servers. Retail investors are competing against highly sophisticated, millisecond-fast automated strategies they cannot access. A July 2025 SEBI interim order banned Jane Street, an American quantitative trading firm, for index manipulation, highlighting how high-frequency traders exploit retail participants in the F&O market. Furthermore, individual traders focus purely on whether an option premium moves up or down, completely ignoring the massive layer of statutory taxes, platform fees, and transaction charges applied to every click. Over 70% of these transaction costs stem from brokerage and exchange fees. In FY24, retail traders incurred a total net loss of ₹74,800 crore. Of this, ₹22,450 crore went entirely toward transaction costs. When you strip away these costs, individual traders suffered a gross trading loss of ₹52,400 crore. Crucially, this wealth was transferred directly to the other side of the aisle: proprietary traders booked ₹33,000 crore in gross profits, while Foreign Portfolio Investors (FPIs) took home ₹28,000 crore. Both SEBI and the Union Government have tried to cool this market. Regulators have hiked minimum contract sizes, limited weekly expiries, mandated upfront premiums, enforced a 50% cash collateral rule, introduced stringent intra-day position monitoring and forced brokers to display prominent risk disclosures. While these initiatives caused a brief contraction in overall F&O volumes, retail investors are stubbornly staying put. In fact, NSE volumes from

May 2026 show that the proportion of individual investors in the Equity Derivatives pool actually rose by 4% year-on-year to hit 31%. As FPIs and proprietary traders reduce their exposure, retail traders are choosing to make up a larger slice of a highly volatile, shrinking pie.

The way forward

Indian regulators must now look toward global safeguards. Mature derivatives markets do not rely on warnings; they enforce strict entry barriers and suitability systems to limit speculative risks.

For example, in Singapore, the Monetary Authority of Singapore requires brokers to conduct a formal Customer Knowledge Assessment for exchange traded derivatives. Similarly, in the U.S., under the Financial Industry Regulatory Authority's Know Your Customer and Suitability Rules, broker-dealers act as legally liable gatekeepers. Brokers must bucket retail clients into strict options approval tiers based on their self-certified financial health; or face severe fines and civil liabilities if they approve unseasoned traders for complex strategies.

Therefore, to protect vulnerable citizens, SEBI must move beyond passive warnings. This regulatory courage is especially critical as the National Stock Exchange (NSE) prepares for its long-awaited public debut. With a massive portion of the NSE's IPO valuation anchored directly to the explosive transaction volumes of retail F&O trading, the exchange's commercial incentives are fundamentally at odds with investor protection. SEBI must see past the market euphoria.

The regulator should explore implementing a baseline liquid capital threshold for F&O access, establish mandatory entry exams, and place the legal responsibility of verifying investor suitability squarely on the shoulders of the discount brokerages profiting from the volume, ensuring that a public listing for the NSE does not come at the cost of financial stability.

India's F&O Boom Needs Adequate Protections

Derivatives are financial contracts, set between two or more parties, that derive their value from an underlying asset, group of assets, or benchmark.

- The underlying assets can be Shares, Bonds, Currencies, Commodities etc
- They are used to hedge (or protect) risks arising from future price fluctuations.
- They can be exchange traded or can be outside exchange (known as Over the Counter or OTC).
- They are regulated by SEBI.

Forwards: A forward contract is a customized contract between two parties to buy or sell an asset at a specified price on a future date. These happen outside the stock exchange and are known as Over the Counter Contracts.

Futures: A futures contract is a standard agreement to buy or sell a particular commodity asset, or security at a predetermined price at a specified time in the future. Futures contracts are standardized for quality and quantity and are traded on future exchange platforms.

Options: An options contract gives the holder the right to buy or sell an underlying security at a predetermined price, known as the strike price. It is not the obligation of the option buyer to perform the contract.

India's F&O Boom Needs Adequate Protections

Concept

- The Problem

- India's Futures and Options (F&O) trading boom has led to devastating financial losses for retail investors, with mounting debt pressure recently linked to multiple suicide cases

- Scale of Retail Losses

- A July 2025 SEBI study found over **90% of individual traders incurred heavy losses over FY22-FY25**; aggregate retail losses surpassed Rs 2.8 lakh crore, **with only 1% of retail traders making over Rs 1 lakh profit**

- The "Rigged Playing Field"

- **97% of institutional profits and 96% of proprietary trading profits come from algorithmic trading using exchange-co-located code** retail traders compete against sophisticated systems they cannot access

India's F&O Boom Needs Adequate Protections

- **What Are Futures and Options?**

- Derivative contracts whose value is derived from an underlying asset; futures obligate a buy/sell at a future date, options give the right (not obligation) to do so at a set price

- **Why Institutions Use Derivatives**

- Primarily to hedge existing portfolios against market risk, unlike many retail participants who use them for speculative, leveraged bets on short-term price movements

- **Algorithmic and High-Frequency Trading**

- Automated trading using computer algorithms to execute orders at extremely high speeds, often co-located within exchange infrastructure for latency advantages

- **Global Regulatory Models**

- Singapore's Monetary Authority requires formal Customer Knowledge Assessments; the US's FINRA enforces Know-Your-Customer and Suitability Rules making brokers legally liable gatekeepers

India's F&O Boom Needs Adequate Protections

58) Consider the following statements:

- I. India accounts for a very large proportion of equity option contracts traded globally thus exhibiting a great boom.
- II. India's stock market has grown rapidly in the recent past even overtaking Hong Kong's at some point of time.
- III. There is no regulatory body either to warn the small investors about the risks of options trading or to act on unregistered financial advisors in this regard.

Which of the statements given above are correct?

a. I and II only

b. II and III only

c. I and III only

d. I, II and III

Making Sense of Embodied AI: The Next Frontier in Robotics

Making sense of embodied AI: the next frontier in robotics

Unlike the AI chatbots that learn from texts, images, and videos, embodied AI must master gravity and balance, which are more difficult due to the countless physical scenarios a robot must face and the simulation-to-real gap, where success in simulation rarely translates perfectly to the real world

John Xavier

In April 14, Boston Dynamics announced that Spot, the yellow-colour robot dog that spent most of its life running narrowly scripted routines, is getting an AI brain. Gemini Robotics-R 1.6 is being folded into Spot and into Boston Dynamics' Orbis inspection platform, giving both stronger spatial reasoning, autonomous decision-making, and the ability to keep learning inside complex industrial sites. Spot's hardware has been good for years, but the intelligence needed to make it useful in a messy factory, without a human driving it frame by frame, is only arriving now. That gap – between a machine that can move and a machine that understands what it's moving through – is exactly what a decades-old field called embodied AI has been trying to close.

What 'embodied' actually means
The easy mistake is to think embodied AI just means "AI in a robot body." The deeper claim by experts working on this field is that a body is not a delivery mechanism for intelligence, but a part of the computation itself.

In their book *How the Body Shapes the Way We Think*, Rod Pfeifer, longtime director of the Artificial Intelligence Laboratory at the University of Zurich, and Josh Bongard, computer science professor at the University of Vermont, argue that thought is not independent of the body but is tightly constrained, and at the same time enabled, by it.

Pfeifer has worked extensively on building physical robots to test behaviours that emerge from the body's design and look for results that tell something about intelligence in general.

Australian roboticist Rodney Brooks, who wrote the foreword for the book and whose own work at MIT in the late 1980s and 1990s made the case even more provocatively, asserting against the dominant symbolic-AI paradigm of the time, that robots didn't need internal world models and elaborate planning to act intelligently. A layered "subsumption architecture" of simple reflexes, tightly coupled to sensors and motors, could produce robust, real-time behaviour without anyone programming a representation of the world at all.

You don't need to build an abstract map of a room if your sensors and body can just react to the room as it actually is, moment to moment. Practically, this plays out as "morphological computation" – offloading work that would otherwise require a brain onto the physical structure of the body.

A passive-dynamic walker can stroll down a slight incline with no motors or control system at all, purely because the leg geometry and joints are shaped to do the computing. A soft, compliant robot hand can grasp an oddly shaped object without a controller that has explicitly modelled that shape, because the material itself deforms and adapts.

A common thread connecting Pfeifer's Zurich lab, Brooks's robots, and today's humanoid is that intelligence is distributed between brain, body, and environment, not confined to just one part.



Boston Dynamics' Atlas robot makes a yoga pose on the football field. SPECIAL ARRANGEMENT

Why this matters now, and why it's hard

The current wave of interest is precisely about discovering how hard that distribution problem is at scale. Unlike the AI chatbots that learn from texts, images, and videos, embodied AI must master gravity and balance, which are profoundly difficult because of the countless physical scenarios a robot must face and the "simulation-to-real gap," where success in simulation rarely translates perfectly to the real world.

Industry data bears this out starkly. Though the embodied AI market is projected to reach \$23 billion by 2030, most humanoid robots still last only about 90 minutes on a charge, and policies that succeed 95% of the time in the lab drop to roughly 50% in the real world. The gap between demo and deployment is the field's central, unglamorous problem.

Embodied AI vs. neuromorphic AI
In common discussions, people tend to conflate embodied AI with neuromorphic AI as both take biology seriously. But they answer different questions.

Embodied AI is about where intelligence lives. The field views cognition distributed across brain, and body, meaning computational work is not fixed to the brain. It is largely agnostic about what kind of processor sits inside the robot; you could run an embodied robot's "brain" on an ordinary GPU cluster.

Neuromorphic AI is about how the processor itself is built – it is a hardware and algorithm paradigm that tries to mimic the physical mechanics of biological neurons, most commonly through spiking neural networks (SNNs).

SNNs operate through discrete spikes; each neuron integrates incoming signals and fires only when a threshold is exceeded, which makes them well suited to time-sensitive tasks like motion

sensing. Neuromorphic chips are also notably power-efficient, since only the neurons actively spiking consume energy at any given moment.

The two areas in robotics increasingly overlap in practice as a growing body of research on "embodied neuromorphic intelligence" puts spiking, event-driven chips inside physical robots specifically because their low power draw and timely response suit real-world, always-on embodied tasks.

Where evolution comes in

If bodies do computational work, an obvious question follows: how do you design the right body for the task, instead of bolting an AI model onto whatever frame engineers have built?

This is the terrain of evolutionary computation.

Yaochu Jin, an Alexander von Humboldt Professor at Bielefeld University, makes the case that neural control and physical form should not be designed separately and then combined; instead, the two have to be developed together, the way biological organisms grow nervous systems and bodies in tandem.

Mr. Jin's research sits at the intersection of evolutionary algorithms, developmental biology, and computational neuroscience, focused on the interaction between evolution, learning, and artificial systems.

His research focuses on co-evolving nervous systems and morphology, and modelling how feedback from the environment shapes an organism's sensory distribution.

This matters directly for embodied AI's central bottleneck: robot bodies are frequently mismatched to the tasks and environments they end up in, and manually re-engineering hardware every time a task changes is slow and expensive. Evolutionary methods offer a way to

search body and brain design space automatically, letting simulated populations of robotic forms compete and replicate based on task performance before anything is built.

The unresolved problems

Even as capital and models pour into the field of embodied AI, a fairly consistent list of hard problems keeps surfacing. For instance, the gap between simulation and real-world experience in using robotic systems is still not narrow. Policies trained in simulation that can be run millions of times cheaply, degrade sharply when deployed onto real hardware.

Sophisticated reasoning models are often too slow to keep pace with a robot arm or leg that needs to react in milliseconds, forcing a split between heavy "thinking" done off-device and lighter reflexive control on it. Short runtimes due to extensive battery use and vulnerable components keep undercutting otherwise successful pilots.

These systems also lack data. Text and image models were scaled on troves of internet data; embodied systems have no equivalent. Efforts like the Open X-Embodiment dataset and Generalist AI's GEN-0, pretrained on hundreds of thousands of hours of manipulation data, are attempts to build that missing corpus, but to put these systems in real-world situations they need to be trained on at least tens of millions of hours of data.

Embodied AI should be viewed as a systems challenge rather than only a software challenge as safe deployment depends on sensors, hardware robustness, operational design limits, human interaction, cybersecurity, and organisational processes, not algorithms alone. Regulators must start asking for evidence of what counts as proof for allowing learning-enabled robotics systems to be deployed in real world.

All of these problems cannot be solved by a single breakthrough. They're being chipped away in parallel with better simulators to narrow the simulation-to-real gap, larger shared datasets to address data scarcity, split "brain in the cloud, reflex on the robot" architectures to handle the speed mismatch, and slowly maturing safety frameworks to handle governance.

Evolutionary methods of the kind Mr. Jin champions add another angle to that effort.

Instead of only improving how a fixed body learns, they ask whether a different form factor would make the problem easier to solve in the first place.

Embodied AI systems have come a long way. Boston Dynamics' humanoid robot Atlas on the football field during FIFA World Cup 2026 shows how robots are able to adapt to uneven surfaces. In a separate event, on America's Got Talent Season 21, China's Ustream GI robots performed alongside Wu Yufei, a 26-year-old Chinese dancer. Spot is now powered by an AI brain and can assist e-commerce delivery agents.

It is starting to become clear that intelligence was never just a matter of better software sitting inside better computers. It was also, at least partly, a matter of testing out new form factors. And these designs are starting to get better with advances in quadruped and avian design-inspired robots.

Making Sense of Embodied AI: The Next Frontier in Robotics

- **Embodied AI is AI that has a physical body and can perceive, move, and interact with its environment to perform tasks autonomously.**
- **Neuromorphic AI is AI that is designed to work like the human brain by mimicking the way neurons and synapses process information.**

India, Spain Advance Talks on UPI-Bizum Interoperability

India, Spain advance talks on UPI-Bizum interoperability

New Delhi: India and Spain have agreed to fast-track discussions on enabling interoperability between India's Unified Payments Interface (UPI) and Spain's Bizum digital payments platform, the Commerce Ministry said on Sunday.

The two countries have also discussed ways to strengthen investment facilitation, improving mobility of professionals and promoting joint investments in Latin America, leveraging Spain's strategic position as India's gateway to Europe and Latin America.

These issues came up for discussion during the recent visit of Commerce and Industry Minister Piyush Goyal to Spain.

"Both sides also agreed to advance technical discussions for interoperability between India's UPI and Spain's Bizum digital payments platform," he said. The two sides have also identified key opportunities in automobiles, pharmaceuticals, chemicals, textiles, renewable energy, tourism and infrastructure. PTI

India, Spain Advance Talks on UPI-Bizum Interoperability

- **Digital Public Infrastructure** refers to a collection of shared digital platforms and systems that operate at the scale of an entire society. These systems are designed to be secure, interoperable, and built on open standards, enabling both public and private entities to provide services efficiently.
- **UPI is an advanced version of Immediate Payment Service (IMPS)**. It is a round-the-clock real-time payment system that facilitates inter-bank peer-to-peer (P2P) and person-to-merchant (P2M) transactions.
- **It was developed by the National Payments Corporation of India (NPCI),** an umbrella organization for operating retail payments and settlement systems, initiated by the **Reserve Bank of India (RBI)** and the **Indian Banks' Association (IBA)**. It is a **“Not for Profit” Company under Section 8 of Companies Act 2013.**
- It has transformed India **“from queues to QR codes (Quick Response codes),”** emerging as the backbone of the country’s digital financial ecosystem.
- **BHIM (Bharat Interface for Money):** It is a mobile payment application developed by the NPCI that allows users to send and receive money instantly using the UPI.
- **Global Acceptance :** Bhutan, France, Mauritius, Nepal, Singapore, Sri Lanka, UAE, Qatar and Cambodia
- **UPI has 2 factor authentication**

India, Spain Advance Talks on UPI-Bizum Interoperability

MDR is a fee charged for the merchants by the bank for accepting payments from customers through credit/debit cards/QR Code in their establishments.

In December 2019, the Government decided to waive off MDR charges on transactions done through RuPay and BHIM-UPI payments to push digital payments. This came into effect from Jan 1, 2020.

India, Spain Advance Talks on UPI-Bizum Interoperability

Q. Consider the following countries :

I. United Arab Emirates

II. France

III. Germany

IV. Singapore

V. Bangladesh

How many countries amongst the above are there other than India where international merchant payments are accepted under UPI?

(a) Only two

(b) Only three

(c) Only four

(d) All the five

India, Spain Advance Talks on UPI-Bizum Interoperability

With reference to digital payments, consider the following statements:

- 1. BHIM app allows the user to transfer money to anyone with a UPI-enabled bank account.**
- 2. While a chip-pin debit card has four factors of authentication, BHIM app has only two factors of authentication.**

Which of the statements given above is/are correct?

This question was previously asked in
**Official UPSC Civil Services Exam
2018 Prelims Part A**

[Download PDF](#)

[Attempt Online](#)

[View all UPSC Civil Services Papers >](#)

1. 1 only
2. 2 only
3. Both 1 and 2
4. Neither 1 nor 2

India, Spain Advance Talks on UPI-Bizum Interoperability

Which one of the following best describes the term "Merchant Discount Rate" sometimes seen in news?

This question was previously asked in
**Official UPSC Civil Services Exam
2018 Prelims Part A**

[Download PDF](#)

[Attempt Online](#)

[View all UPSC Civil Services Papers >](#)

1. The incentive given by a bank to a merchant for accepting payments through debit cards pertaining to that bank.
2. The amount paid back by banks to their customers when they use debit cards for financial transactions for purchasing goods or services.
3. The charge to a merchant by a bank for accepting payments from his customers through the bank's debit cards.
4. The incentive is given by the Government to merchants for promoting digital payments by their customers through Point of Sale (PoS) machines and debit cards.

Thank You